

LEVERAGING ERROR-TOLERANCE ASYMMETRY IN AUTOMATED ELECTRICAL GRID VISUAL INSPECTION WITH A SEMI-SUPERVISED ANNOTATION PIPELINE

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ABSTRACT

Annotating large-scale visual datasets for electrical grid inspection is labor-intensive, particularly if exhaustive labeling of all assets is required. However, for defect detection tasks, labeling errors in non-defective assets are more tolerable than errors on defective instances. We explore this error-tolerance asymmetry to propose a semi-supervised annotation pipeline that efficiently constructs high-quality training datasets from partial supervision. Given limitations on the available human experts and consequently on annotation resources, we propose labeling defects across thousands of images, while remaining assets can be labeled in only a small subset, used to train an asset detector that further scales asset-level annotations. Human and model-generated labels are then merged, using a supervision-aware strategy that prioritizes defect annotations. To address the challenges of long-distance image acquisition while avoiding downsampling high-resolution inspection imagery, supervision-aware image cropping is applied to focus training on regions of interest. Finally, coarse bounding boxes are refined into accurate polygonal instances using visual prompting with large vision models, while preserving supervised class identities. Ablation studies with state-of-the-art models demonstrate the effectiveness of each stage of the proposed pipeline.

Index Terms— Annotation pipeline, Labeling efficiency, Semi-supervised, Error-tolerance asymmetry, Electrical grid.

1. INTRODUCTION

The increasing reliance on electrical power is demanding for efficient maintenance to ensure grid reliability. Visual inspection, increasingly scaled through UAV-based remote data collection, remains essential. While deep vision models have the potential to automate inspections, current systems still underperform human experts, often due to dataset limitations [1].

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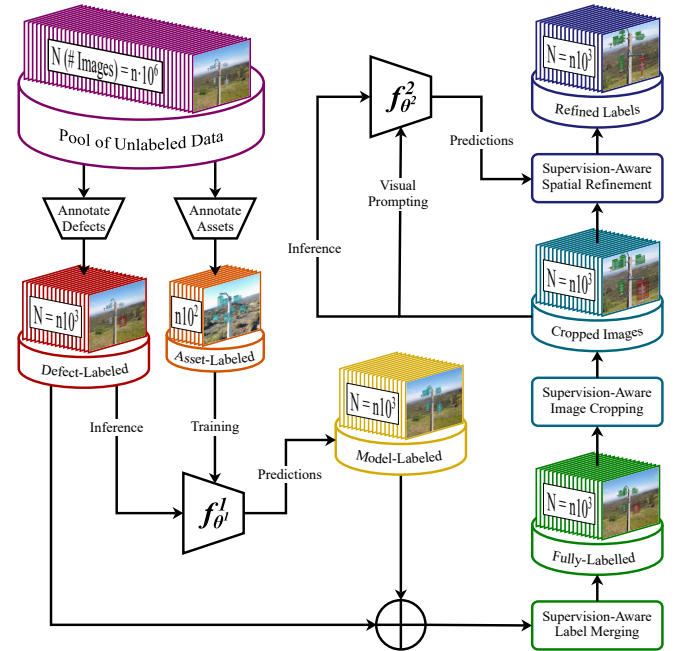


Fig. 1: Overview of the proposed supervision-aware annotation pipeline. Starting from unlabeled data, supervision through human annotations is obtained at different scales for defects and assets. An initial model f_{θ}^1 , trained on asset-labeled data, generates model-predicted annotations. Defect and asset labels are then merged using a supervision-aware function, that prioritizes human labels, producing a fully labeled dataset. Next, supervision-aware image cropping is applied to “zoom in” on regions of interest. Finally, annotations spatial detail is refined via visual prompting with model f_{θ}^2 , while preserving the supervised class identities.

Although inspection imagery collected for human assessment can—and should—be used in automated analysis, data requirements of each method have fundamental differences.

Modern UAV are equipped with high-resolution sensors to enable long-range inspection and reduce operational risk. Human inspectors can naturally and efficiently interpret such imagery by dynamically zooming into regions of interest, focusing attention where needed with minimal effort.

Deep learning architectures lack this adaptive capability. Processing native 48MP imagery is computationally impractical, while aggressive downsampling risks discarding small but critical visual cues. As a result, efficient data processing requires a strategy to mimic human zooming behavior by focusing computation on more relevant regions of the image.

Image cropping provides a natural way to focus on regions of interest, but to be done effectively, all assets should be detectable. Detecting assets, such as voltage insulators, is relatively straightforward due to their visual distinctiveness, even though it provides limited operational value on its own. In contrast, detecting defects is substantially more challenging: defective and non-defective assets can appear very similar, yet accurately identifying defects is critical for inspection outcomes. Operational inspection workflows typically produce sparse annotations. Human inspectors focus on defective components to support maintenance decisions and do not waste resources on exhaustively label non-defective assets. Moreover, requiring precise geometric annotations, such as tight bounding boxes or polygons, further increases labeling effort without direct benefit for inspection reporting.

The mismatch between operational labeling practices and object detection training requirements is a key contributor to the scarcity of annotated datasets. Sparse annotations provide limited supervision for training. When only defects are labeled, non-defective assets are treated as background, reducing informative gradient signals and slowing learning. The lack of contextual supervision hinders discrimination between subtle defects and visually similar non-defective components, biases the model toward the labeled subset, and increases false positives. Including approximate annotations for non-defective assets provides contextual cues, stabilizes gradients, and improves training efficiency, particularly in high-resolution images where defects occupy a small fraction of pixels. Fine-grained spatial annotations, such as tight bounding boxes or polygon masks, further strengthen localization signals and gradient quality, especially for small or densely packed objects. At test time, exact spatial alignment is often unnecessary: defects detected with approximate localization remain operationally actionable. Accordingly, evaluation metrics should tolerate reasonable spatial deviations to avoid underestimating performance.

Not all annotation errors have equal impact—accurate semantic labeling of defective instances is critical for both training and evaluation, whereas minor inaccuracies in asset instance labels are acceptable, as they primarily provide context and support cropping rather than drive inspection decisions.

This error-tolerance asymmetry motivates a supervision-aware annotation pipeline. The proposed method is presented in Fig. 1, in which human supervision effort is focused on defects, while label quantity of non-defective instances is leveraged by model predictions, that are further used to support dynamic image cropping and effective training without increasing annotation burden or affecting operational performance.

1.1. Key Contributions

- **Error-tolerance-aware formulation.** We formalize the asymmetric impact of annotation errors in defect detection, where accurate defect labels are essential while approximate non-defective asset labels are sufficient for supporting effective training.
- **Supervision-aware dataset construction.** We propose a practical pipeline that combines partial human supervision with model-generated labels, labeling spatial refinement, and annotation-driven cropping.
- **Operationally aligned evaluation.** We evaluate defect detection using decomposed IoU metrics (IoR, IoP), which separately measure coverage and precision, providing robustness to realistic annotation noise and better reflecting operational inspection requirements.
- **Experimental validation.** We demonstrate consistent gains in detection performance through ablation studies using a state-of-the-art detector over multiple runs.

2. RELATED WORK

Progress in deep learning has been closely tied to the availability of annotated datasets. In computer vision, advances in model performance have historically followed increases in dataset size and annotation richness. Since dense supervision is expensive, recent work has explored weakly supervised, semi-supervised, and data-centric strategies that reduce labeling effort while preserving model effectiveness.

2.1. Generic Domain Datasets

Introduced in 1998, the MNIST dataset [2], reveal deep learning potential for optical character recognition through a 70k grayscale images dataset of handwritten digits. The introduction of ImageNet [3] in 2009, marked another major shift, by scaling visual recognition to 1,000 classes across over 1.2 million images, showing that large-scale labeled data is critical for training high-capacity models. Subsequent datasets expanded both task complexity and annotation richness. MS-COCO [4], released in 2015, introduced object detection and instance segmentation benchmarks with roughly 330k images and over 1.5 million annotated instances, while in 2019 LVIS [5] increased category diversity to over 1,200 long-tailed classes, highlighting the growing cost and complexity of dense supervision. More recently, dataset construction has shifted toward scale and efficiency. FLD-900M [6], introduced in 2021, provided hundreds of millions of image-text pairs to support multimodal learning, reducing reliance on dense spatial annotations. In a similar direction, the SA-1B dataset [7] released in 2023 demonstrated how automated annotation loops can generate over one billion segmentation masks using limited human supervision, emphasizing the role of model-assisted data collection in scaling visual datasets.

2.2. Electrical Grid Datasets

Domain-specific inspection datasets have generally lagged behind generic vision benchmarks. Proprietary data, privacy concerns, and safety regulations limit dataset sharing, slowing collective progress. Limited annotation budgets and human labeling resources further restrict dataset scale and richness.

The Electric Power Research Institute (EPRI) has made efforts to advance automated inspection through public dataset releases [8], including over 40k UAV-acquired images of transmission [9] and distribution [10] systems, annotated across multiple asset classes and useful for inventory and structural analysis. Another EPRI dataset [11] containing ground-based images from foot inspections, with labels for healthy, broken, and flashover-damaged insulators. While these datasets support defect detection, they are often biased toward close-range, tightly framed imagery and do not fully reflect the long-range conditions typical of UAV inspections.

Fortunately, the widespread adoption of UAV-based remote inspections is transforming operational practices by enabling large-scale data collection and rapidly increasing available imagery. Although these workflows are primarily designed for anomaly reporting—which typically occurs in less than 1% of the data—existing inspection datasets with partial annotations can be adapted for efficient defect detection. By accounting for error-tolerance asymmetry and leveraging model-assisted annotation, these datasets can be extended without requiring impractical levels of manual labeling, motivating new strategies for dataset construction.

2.3. Literature Gap

Most studies, particularly in applied domains such as electrical grid inspection, adopt a model-centric approach, focusing frequently on *doing the thing right* rather than *doing the right thing*. This misalignment can manifest as dataset limitations, inconsistent validation practices, and a divergence between metric optimization and operational performance [1].

Dataset construction strategies that perform well for generic vision tasks often struggle when applied to industrial inspection data. Inspection datasets face unique challenges that standard benchmarks do not: defects are sparse, images are high-resolution, and annotation budgets are limited.

Data-centric approaches have shown modest gains from improving annotation spatial detail, suggesting that finer localization strengthens training signals [12]. However, strict precision requirements at test time can mask these benefits. Using lower IoU thresholds or decomposed metrics like IoR and IoP can better capture the impact of spatial detail while aligning evaluation with operational needs.

Existing approaches rarely consider the asymmetric cost of annotation errors between defective and non-defective assets, motivating supervision-aware labeling strategies tailored to defect-focused inspection tasks.

3. METHODOLOGY

This work adopts a data-centric approach to improve computer vision performance through targeted dataset construction. We propose a supervision-aware annotation pipeline that incrementally enhances training data along multiple dimensions, including annotation coverage, spatial accuracy, and effective use of high-resolution imagery. Each dataset variant, illustrated in Fig. 2, is used to train a state-of-the-art object detector, enabling evaluation of the contribution of each data improvement to learning stability and detection performance.

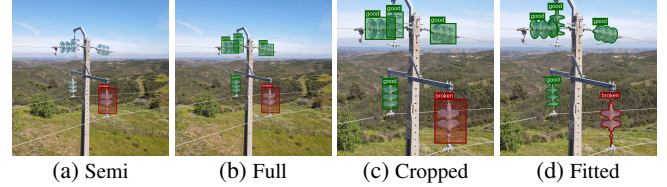


Fig. 2: Canonical examples illustrating successive dataset variants. Starting from defect-only supervision, asset annotations are scaled through model-assisted labeling, followed by supervision-aware cropping to focus on regions of interest, and finally refined into polygonal instances using model-assisted spatial localization.

Data Source: The dataset consists of high-resolution RGB images (24–48 MP) collected by EDP Labellec [13] during UAV-based electrical grid inspections conducted between 2023 and 2025, capturing substantial variability in viewpoint, background, and illumination. As illustrated in Fig. 1, these images cover three levels of supervision, from millions of unlabeled images (purple) to thousands of defect-labeled (red) and hundreds of asset-labeled images (orange). The defect-labeled subset contains 1,835 images with 1,952 broken insulators (≈ 1.1 instances/image), while the asset-only subset comprises 396 images with 2,741 fully annotated functional insulators (≈ 6.9 instances/image). To stratify the data by capture distance, we compute the mean of the square roots of all bounding box areas in an image divided by the square root of the image area. This provides a linear measure of the fraction of the image occupied by an object, which is more interpretable than using quadratic area units. Images are then sorted by this metric and divided into ten bins. To mitigate extreme outliers (see Fig. 3), the 2nd and 9th bins are reserved as close-range and distant test sets, while the remaining bins are used for training under a 4-fold cross-validation.

Model-Assisted Labeling: As illustrated in Fig. 1, an insulator detector was trained on the small asset-labeled subset (orange) and used to generate pseudo-labels for the larger defect-labeled set (red to yellow). The 396-image, 2,741-instance dataset was split into five folds, with one fold held out for testing and the remaining folds used for cross-validation. The model achieving the highest test AP_{50} (0.966) was applied to detect insulators across the 1,835-image defect-labeled dataset. Inference used Non-Maximum Suppression (NMS) with a 0.25 IoU threshold and 0.5 confidence threshold, yielding 14,742 pseudo-labeled assets.

Supervision-Aware Label Merging: With model-labeled data available (yellow), we next integrate them with human annotations (red) using a supervision-aware merging step (green), in which human annotations are preserved. Model-predicted asset boxes overlapping human-labeled defects with $\text{IoU} \geq 0.3$ are removed, as these typically correspond to false positives. Partial overlaps below this threshold are retained, allowing approximate labels for other assets while accounting for the natural overlap of electrical components in a 3D projection. To suppress other common failures, such as multiple nested predictions, we apply an Intersection-over-Prediction (IoP) suppression rule with a threshold of 0.7, which proved more effective than standard IoU-based NMS in this setting. This procedure flagged 183 defect annotations without a corresponding prediction, 28 cases of multiple overlapping predictions, and 158 redundant detections, out of 14,742 total instances (1,952 defective, 12,790 non-defective). Overall, model-generated asset labels miss less than 10% of defective instances, and since defective instances account for less than 15% of all objects, this residual error ($\approx 1.5\%$) is consistent with expected detector performance and acceptable for its role as contextual supervision in defect detection.

Supervision-Aware Image Cropping: To avoid inefficient processing of sub-sampled high-resolution images, we apply supervision-aware cropping to remove irrelevant background while preserving all annotated content. For each image, we compute the tightest axis-aligned region enclosing all asset and defect annotations, expand it by a 20% padding margin to retain nearby context, and extract a square crop compatible with standard detection architectures. If the padded region exceeds image boundaries, the crop center is shifted to ensure a valid square fully contained within the original image. All labels are transformed accordingly to the scaling and translation factor. Empirically, only 3 out of 1,835 images ($\approx 0.16\%$) have no assets detected, confirming that the probability of excluding a defective insulator after cropping is negligible. This approach can be extended to deployment by applying a lightweight asset detector as a preprocessing stage.

The YOLO26 [14] detection heads operate at strides of 8, 16, and 32 (P3–P5), with theoretical receptive fields spanning roughly 10%, 20%, and 40% of the image side at 1280×1280 resolution. However, effective receptive fields are smaller due to Gaussian decay of influence across layers [15]. Features below a few percent of the image side become increasingly diluted. While full insulators align well with these regimes, defect cues—often an order of magnitude smaller—approach the resolution limit, yielding weaker representations.

Figure 3 shows cropping effects on object scale. Original dataset insulators mean normalized size occupy $\mu = 0.062$ ($\sigma = 0.021$) of image side, placing defects near or below the P3 effective field. Cropping increases this to $\mu = 0.117$ ($\sigma = 0.035$), doubling relative scale and better aligning defects with the detection heads receptive fields, which are critical for fine-grained defect localization.

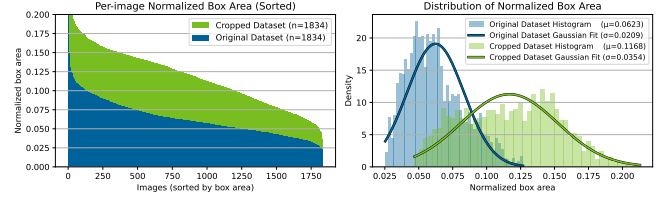


Fig. 3: Normalized object scale distributions before and after supervision-aware cropping. Cropping increases relative object size while preserving scale variability essential for multi-scale training.

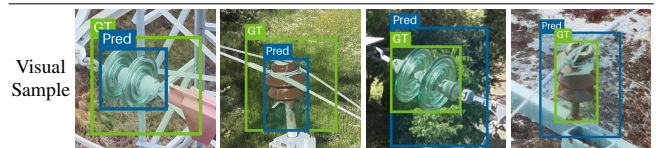
Supervision-Aware Spatial Refinement: We use original bounding boxes as visual prompts for SAM3 [16] to generate polygonal masks that are further constrained to the box extent, and refined via morphological closing to suppress artifacts. Only the largest connected component is retained, and boundaries are simplified using the Ramer–Douglas–Peucker algorithm [17]. Predictions are matched to original instances by an $\text{IoU} \geq 0.25$. Of 14,740 instances, 14,640 are matched. 26 unmatched model-labeled *good* instances are discarded, while 74 unmatched human *defect* annotations are reverted to box geometry. Additionally, 154 matched predictions are also reverted to box based on a boundary irregularity criterion (isoperimetric quotient $S = \frac{P^2}{4\pi A} - 1 > 10$). Resulting 14,714 refined labels that preserved all human-provided identities.

Implementation Details: All experiments were performed with identical training protocols, hyperparameters, and evaluation metrics across all dataset variants. Yolo26 [14] medium architecture models are trained for 100 epochs with a batch size of 32 and an input resolution of 1280×1280 , using the MuSGD optimizer ($\text{lr } 7.5 \times 10^{-4}$, momentum 0.9), and 10-epoch warm-up. We apply a dropout rate of 0.1, a weight decay of 0.001, disable hard data augmentations, and double the classification loss weight. To account for variability, models were trained under 4-fold cross-validation and 4 random seeds, with each resulting model evaluated on a separate test set, ensuring that mean performance differences primarily reflect data quality rather than training stochasticity.

Evaluation Metrics: Standard IoU penalizes small localization deviations and true coverage failures equally, though minor offsets matter less when annotations are inconsistent. To avoid common metrics limitations [18], we report IoR and IoP separately, distinguishing under-coverage (missed defects) from over-coverage (annotation imprecision). Table 1 presents failure modes that IoU alone obscures.

Table 1: IoU, IoR, and IoP metrics for representative predictions.

<i>IoU</i>	0.388783	0.390583	0.404719	0.415551
<i>IoR</i>	1.000000	1.000000	0.404719	0.415551
<i>IoP</i>	0.388783	0.390583	0.999996	1.000000



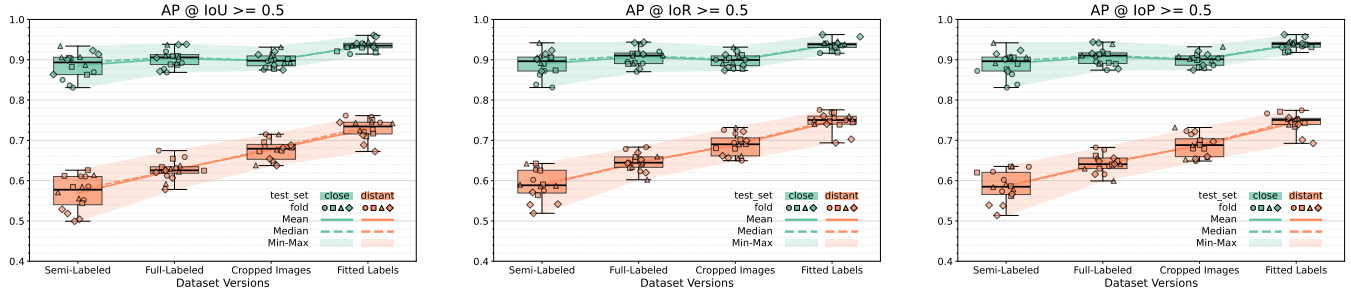


Fig. 4: Detection performance (AP_{50}) for broken-class instances across 128 experiments: 2 test $\times$ 4 datasets $\times$ 4 folds $\times$ 4 seeds.

4. RESULTS AND DISCUSSION

Reported performance is inherently conditioned by test set distribution. We therefore evaluate separately on close (green) and distant (orange) imagery, enabling analysis across difficulty levels rather than relying on a single, potentially biased distribution. Figure 4 presents detection performance, exclusively for broken-class instances, across the four dataset variants, semi-labeled, fully-labeled, cropped images, and fitted labels (see Fig. 2). Each box shows the interquartile range and median across 16 runs (4 folds $\times$ 4 seeds), with individual points overlaid and whiskers extending to minimum and maximum values. Metrics include standard AP_{50} (IoU-based) as well as AP_{50} computed separately for IoR and IoP .

Progressive dataset refinement demonstrated consistent additive gains. For close-range images, $AP_{IoU \geq 50}$ increases from 0.886 ± 0.031 with semi-labeled to 0.936 ± 0.013 with fitted labels. Distant-range performance similarly improves from 0.569 ± 0.041 to 0.729 ± 0.025 . Adding non-defective asset labels increases mean performance and reduces variance, confirming that contextual supervision improves training convergence. Cropping mainly benefits distant images, where small object scale limits signal quality, highlighting the importance of aligning receptive fields with cue size. Polygonal refinement yields consistent additional gains, indicating that finer spatial detail improves gradient quality even when tight localization is not required at test time.

Annotation variability is inherent in multi-annotator workflows, ranging from tightly to coarse. Audits of widely used benchmarks reveal intriguing annotation errors [19]. As models approach human consistency, test-set noise becomes an evaluation ceiling. Accurate predictions that diverge from imperfect ground truth are penalized as errors, and metric gains may reflect alignment with label error rather than genuine model capability. This tension is most acute in model-centric approaches that emphasize incremental benchmark gains without scrutinizing data quality. While the preferred solution is to reduce label noise, appropriate metrics can mitigate its effects. By decomposing spatial agreement into IoR and IoP , we reduce sensitivity to minor spatial discrepancies unlikely to affect operational decisions. Across all experiments, AP_{50} increases from 0.778 ± 0.138 (IoU) to 0.788 ± 0.130 (IoR), revealing approximately 1% of true positives that standard IoU fails to credit.

5. CONCLUSION

Prior work in visual inspection and object detection has largely focused on model architectures, loss functions, or fully supervised datasets, often assuming uniformly accurate annotations across all classes. Semi-supervised and model-assisted labeling approaches have reduced annotation costs by leveraging model predictions, but they typically treat annotation errors symmetrically and rely on strict IoU-based metrics for evaluation. Recent comprehensive reviews of automated inspection systems [1] have emphasized that these assumptions are poorly aligned with industrial practice, highlighting persistent issues related to dataset limitations, annotation noise, and the mismatch between benchmark metrics and operational requirements. However, these analyses remain primarily conceptual and do not provide experimentally validated strategies for handling asymmetric annotation reliability or evaluation robustness in defect-focused settings.

While prior studies explored the benefits of annotation consistency and spatial granularity [12], the present work translates these insights into a practical, supervision-aware pipeline. Specifically, we address the asymmetric tolerance to annotation errors inherent in defect detection with an efficient annotation strategy: human expertise is used for defect labeling, while non-defective asset labels are model-generated solely to provide contextual guidance during training and enable dynamic image cropping. Precise spatial detail is supplied by a segmentation model, reducing the labeling effort required from human experts. Since enforcing exact spatial alignment at test time can misrepresent operational performance, we adopt decomposed metrics (IoR and IoP) that separately quantify coverage and precision, yielding evaluations that are more robust to realistic annotation noise.

The proposed supervision strategy opens new directions: cost-free non-defective labels provide additional context that can be leveraged through supervised contrastive learning [20], potentially enhancing feature discrimination in imbalanced binary datasets. Future work should continue pursuing data-centric approaches, with a focus on automated cleaning of annotation noise [19]. Overall, our results demonstrate that aligning annotation strategy, spatial refinement, and evaluation metrics with the asymmetric operational requirements of defect detection can certainly improve learning stability and final performance without increasing the annotation budget.

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